RF Signal-Based Wearable Devices for Diagnosis of Multiple Diseases: A Comprehensive Review of FMCW, UWB, mmWave, and Microwave Sensing Technologies (2023–2025)

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***Abstract*—Radio frequency (RF) sensing technologies are emerging as transformative tools for non-invasive, continuous health monitoring through wearable devices. Unlike traditional contact-based sensors that suffer from motion artifacts, skin irritation, and single-parameter measurement limitations, RF-based wearables can penetrate clothing and skin, operate without direct contact, and simultaneously extract multiple physiological parameters. This paper presents the first comprehensive review that systematically covers all four major RF sensing modalities—Frequency Modulated Continuous Wave (FMCW) radar, Ultra-Wideband (UWB) impulse radar, millimeter-wave (mmWave) sensing, and microwave imaging—across six disease categories: cardiovascular, respiratory, diabetes, cancer, neurological, and sleep disorders. We adopt a disease-centric organization, comparing RF technologies within each clinical application to identify the most suitable modality for specific diagnostic tasks. Our analysis encompasses 40 studies published between 2023 and 2025, examining signal processing pipelines, machine learning architectures, and clinical validation outcomes. Key findings reveal that FMCW radar achieves the highest accuracy (95–98%) for cardiovascular monitoring, UWB excels in respiratory assessment with sub-millimeter chest displacement resolution, mmWave sensing shows promise for non-invasive glucose monitoring, and microwave imaging provides clinically viable tumor detection with 89–94% sensitivity. We identify critical challenges including motion artifact suppression, cross-subject generalization, and regulatory pathways, and propose a roadmap for clinical translation of RF wearable diagnostics.**

***Index Terms*—RF sensing, wearable devices, FMCW radar, UWB, mmWave, microwave imaging, health monitoring, disease diagnosis, non-invasive sensing, vital signs, machine learning**

I. Introduction

CHRONIC diseases account for approximately 74% of global deaths annually, with cardiovascular disease, cancer, respiratory disorders, and diabetes collectively responsible for over 35 million fatalities worldwide [1]. The escalating burden of these conditions has intensified demand for continuous, non-invasive monitoring technologies that enable early detection, timely intervention, and long-term disease management outside clinical settings. Wearable health devices have emerged as a cornerstone of this paradigm shift, with the global market projected to exceed $130 billion by 2025 [2].

Traditional wearable sensors—including photoplethysmography (PPG), electrodermal activity (EDA), and electrocardiography (ECG)—have demonstrated clinical utility but suffer from fundamental limitations. Contact-based electrodes cause skin irritation during prolonged wear, motion artifacts degrade signal quality during daily activities, and most sensors target a single physiological parameter, necessitating multiple devices for comprehensive health assessment [22]. Furthermore, optical sensors like PPG exhibit reduced accuracy in individuals with darker skin pigmentation and are sensitive to ambient light interference [28].

Radio frequency (RF) sensing technologies offer a compelling alternative that addresses these limitations. RF signals in the 3–77 GHz range can penetrate clothing, skin, and superficial tissue, enabling non-contact measurement of cardiac motion, respiratory displacement, blood glucose permittivity changes, and tissue dielectric anomalies associated with tumors [30]. A single RF transceiver can simultaneously extract heart rate, respiration rate, and body movement, reducing device complexity while improving user comfort.

Despite the rapid growth of RF-based health sensing research, existing reviews have focused narrowly on individual technologies (e.g., mmWave radar [21]) or single disease applications (e.g., vital sign monitoring [3]). **No comprehensive review currently covers all four major RF sensing modalities—FMCW, UWB, mmWave, and microwave—across multiple disease categories.** This gap limits researchers' ability to identify the optimal RF technology for specific clinical applications and hinders cross-pollination of techniques across disease domains.

This review addresses the following research questions:

- **RQ1:** Which RF sensing technology is most effective for diagnosing each disease category, and what are the performance trade-offs?
- **RQ2:** What signal processing pipelines are employed for each RF modality, and how do they differ across clinical applications?
- **RQ3:** What machine learning and deep learning architectures achieve the highest diagnostic accuracy for RF-based health sensing?
- **RQ4:** What are the critical challenges and gaps preventing clinical deployment of RF wearable diagnostics?

The principal contributions of this paper are:

1) The **first unified review** covering FMCW, UWB, mmWave, and microwave sensing technologies across six

disease categories (2023–2025).
2) A **disease-centric comparative analysis** that identifies the most suitable RF modality for each clinical application.
3) A systematic taxonomy of **signal processing pipelines** for each RF technology, from raw signal acquisition to diagnostic feature extraction.
4) A comprehensive **ML/DL performance benchmark** comparing traditional and deep learning approaches across RF modalities and diseases.
5) Identification of **open challenges** and a research roadmap for clinical translation, including regulatory considerations.
6) A **cross-technology performance matrix** enabling direct comparison of accuracy, power consumption, and form factor across all RF–disease combinations.

Fig. 1 presents the organizational taxonomy of this review.

II. RF Sensing Technologies Overview

A. Frequency Modulated Continuous Wave (FMCW) Radar

FMCW radar transmits a linearly frequency-modulated chirp signal and measures the beat frequency produced by mixing the transmitted and received signals. The beat frequency f_b is proportional to the target range R:

$$f_b = \frac{2R \cdot B \cdot f_s}{c \cdot T_c} \tag{1}$$

where B is the chirp bandwidth, T_c is the chirp duration, f_s is the sweep rate, and c is the speed of light. For vital sign monitoring, the phase variation $\Delta\phi(t)$ of the beat signal encodes sub-millimeter chest wall displacement caused by cardiac and respiratory activity [4]:

$$\Delta\phi(t) = \frac{4\pi}{\lambda}\left[d_r \sin(2\pi f_r t) + d_h \sin(2\pi f_h t)\right] \tag{2}$$

where d_r and d_h are respiratory and heartbeat displacement amplitudes, and f_r, f_h are the corresponding frequencies. Typical FMCW wearable devices operate at 24 GHz or 60–64 GHz with bandwidths of 1–4 GHz, achieving range resolutions of 3.75–15 cm [8].

B. Ultra-Wideband (UWB) Impulse Radar

UWB radar transmits extremely short pulses ($<$2 ns) across a bandwidth exceeding 500 MHz, typically in the 3.1–10.6 GHz range. The time-domain resolution is given by $\Delta R = c/(2B)$, enabling sub-centimeter range resolution. UWB signals penetrate building materials, clothing, and superficial body tissue with low power spectral density ($<-41.3\,\mathrm{dBm/MHz}$), making them inherently safe for continuous body exposure [10]. The received signal reflects chest wall motion, from which respiratory and cardiac signals are extracted via cross-correlation or matched filtering techniques.

C. Millimeter-Wave (mmWave) Sensing

mmWave sensors operate in the 60–77 GHz band, leveraging FMCW or pulsed modulation with wide bandwidths (up to 4 GHz). The short wavelength ($\sim$4 mm at 77 GHz) enables high angular resolution and sensitivity to small physiological displacements. Modern mmWave sensors (e.g., Texas Instruments IWR1443, IWR6843) integrate MIMO antenna arrays that provide spatial multiplexing for multi-target vital sign separation [9]. The higher atmospheric attenuation at mmWave frequencies limits the sensing range but is advantageous for wearable applications where short-range, high-resolution measurement is desired.

D. Microwave Imaging and Dielectric Sensing

Microwave sensing exploits the frequency-dependent dielectric properties of biological tissues in the 0.5–20 GHz range. Malignant tissue exhibits significantly higher permittivity ($\varepsilon_r \approx$ 50–60) than healthy tissue ($\varepsilon_r \approx$ 5–10) due to increased water content and altered cell morphology [16]. Microwave imaging systems illuminate the target with wideband signals and reconstruct dielectric contrast maps using inverse scattering algorithms. For glucose sensing, microwave resonator sensors detect the small permittivity shift ($\Delta\varepsilon_r \approx 0.1$–0.5) in blood caused by glucose concentration changes [13].

E. Wearable Antenna Designs

The translation of RF sensing to wearable form factors requires antennas that conform to the body surface while maintaining radiation performance. Recent advances include textile-integrated patch antennas on felt and denim substrates [17], flexible polyimide-based slot antennas, and 3D-printed conformal arrays. Key design considerations include specific absorption rate (SAR) compliance, detuning compensation due to body proximity, and bandwidth maintenance under bending.

III. Signal Processing and Data Preparation

The transformation of raw RF signals into clinically meaningful diagnostic features requires multi-stage processing pipelines tailored to each sensing modality. Fig. 3 illustrates the generalized RF health-sensing signal processing chain.

A. FMCW Signal Processing

FMCW processing begins with range-FFT applied to the beat signal to separate targets at different distances. For vital sign extraction, the range bin corresponding to the subject's chest is identified, and the phase history $\phi[n]$ across consecutive chirps is unwrapped. A cascaded bandpass filter separates the respiratory component (0.1–0.5 Hz) from the cardiac component (0.8–2.0 Hz). Advanced techniques include Doppler-FFT for velocity estimation and the arctangent demodulation with DC offset compensation to resolve the null-point problem [4]. Recent work by Chen et al. [6] demonstrated that variational mode decomposition (VMD) outperforms empirical mode decomposition (EMD) for separating intermodulation harmonics between respiration and heartbeat.

B. UWB Signal Processing

UWB processing operates primarily in the time domain. The received pulse train is cross-correlated with a template pulse to extract range profiles. Chest displacement is tracked by monitoring the time-of-arrival (ToA) shift of the reflected pulse,

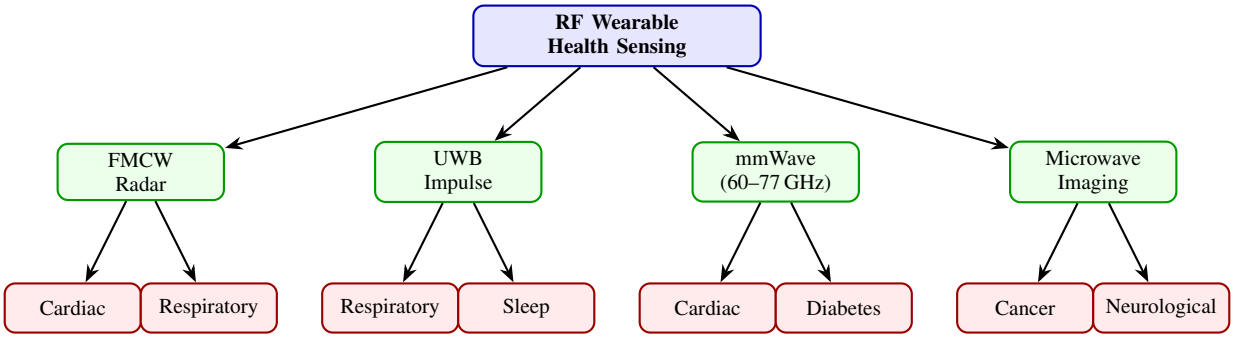

Fig. 1: Taxonomy of RF sensing technologies and their primary disease applications reviewed in this paper. Each RF modality maps to diseases where it demonstrates the strongest diagnostic performance.

TABLE I: Comparison of RF Sensing Technologies for Wearable Health Monitoring

Parameter	FMCW Radar	UWB Impulse	mmWave (60–77 GHz)	Microwave Imaging
Frequency Band	24 / 60–64 GHz	3.1–10.6 GHz	60–77 GHz	0.5–20 GHz
Bandwidth	1–4 GHz	0.5–7.5 GHz	1–4 GHz	0.5–15 GHz
Range Resolution	3.75–15 cm	1.5–30 cm	3.75–15 cm	1–10 cm
Penetration Depth	1–5 mm (skin)	5–30 mm	0.5–2 mm	20–80 mm
Power Consumption	100–500 mW	50–200 mW	150–600 mW	200–1000 mW
Wearable Form Factor	Chest patch, wristband	Chest band, mattress pad	Ear-mounted, patch	Bra-integrated, headband
Body Placement	Chest, wrist, bedside	Chest, bedside, under-bed	Chest, ear, wrist	Breast, head, limb
Contact Required	No	No	No	Near-contact
Multi-target Capability	Yes (MIMO)	Limited	Yes (MIMO)	Yes (array)
Typical Cost	$15–50	$10–30	$20–60	$50–200

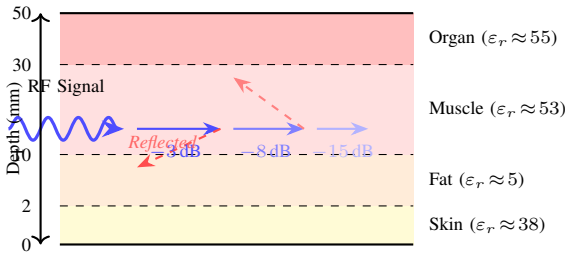

Fig. 2: RF signal propagation through layered human tissue. Signal attenuation and partial reflection occur at each tissue boundary due to dielectric contrast. Permittivity values at 2.4 GHz.

with sub-millimeter resolution achievable through interpolation and averaging [10]. Clutter removal employs exponential averaging subtraction, where the static background estimate is updated as $\hat{B}[n] = \alpha\hat{B}[n-1] + (1-\alpha)S[n]$ with a forgetting factor $\alpha \approx 0.95$. Motion artifact suppression leverages the wideband nature of UWB to discriminate physiological motion (narrowband) from gross body movement (wideband).

C. mmWave Signal Processing

mmWave processing extends FMCW techniques with MIMO-specific enhancements. Multiple-input multiple-output antenna configurations enable angle-of-arrival (AoA) estimation via digital beamforming, permitting simultaneous vital sign monitoring of multiple subjects in the same range bin [9]. The 3D FFT processing chain (range-FFT $\rightarrow$ Doppler-FFT $\rightarrow$ angle-FFT) generates range-Doppler-angle data cubes from which individual targets are resolved. Phase-based vital sign extraction at mmWave frequencies achieves higher sensitivity to small displacements due to the shorter wavelength, but is correspondingly more susceptible to phase noise and random body movement [3].

D. Microwave Signal Processing

Microwave imaging employs S-parameter measurements (reflection S_{11} and transmission S_{21}) across a swept frequency band. For breast cancer screening, the confocal microwave imaging (CMI) algorithm applies time-gating and synthetic focusing to reconstruct a 3D map of dielectric contrasts within the breast tissue [16]. For glucose sensing, the resonant frequency shift Δf_r and quality factor change ΔQ of a microwave resonator are mapped to glucose concentration through calibration curves [13]. Image reconstruction algorithms include delay-and-sum (DAS), MUSIC, and compressed sensing approaches.

E. Feature Extraction Methods

Features extracted from processed RF signals span multiple domains:

- **Time domain:** Mean, standard deviation, root mean square successive difference (RMSSD), peak-to-peak interval, zero-crossing rate.
- **Frequency domain:** Power spectral density (PSD), LF/HF ratio from heart rate variability (HRV), dominant frequency, spectral entropy.
- **Time-frequency domain:** Short-time Fourier transform (STFT) spectrograms, continuous wavelet transform (CWT) scalograms, Wigner-Ville distribution.

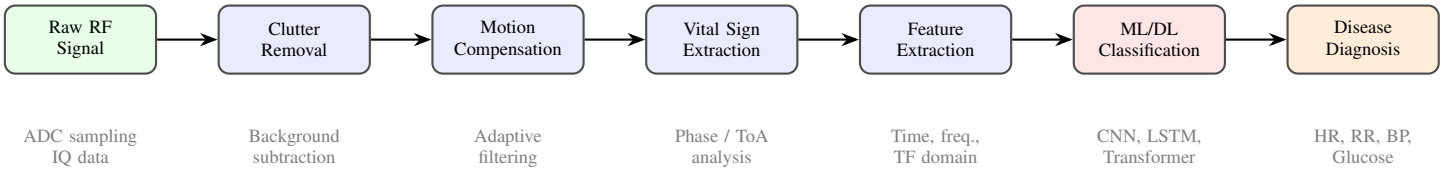

Fig. 3: Generalized RF signal processing pipeline for wearable health sensing. Each stage is adapted based on the specific RF modality and target disease.

- **Learned features:** Convolutional autoencoder latent representations, variational autoencoder embeddings trained on raw RF spectrograms.

IV. Disease-Specific Applications

This section constitutes the core contribution of this review: a disease-centric analysis comparing RF technologies within each clinical domain.

A. Cardiovascular Disease

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide. RF-based cardiac monitoring targets three primary parameters: heart rate (HR), heart rate variability (HRV), and blood pressure (BP).

1) Heart Rate Monitoring: FMCW radar at 60–64 GHz achieves the highest HR accuracy among RF modalities, with mean absolute error (MAE) below 1.5 bpm in controlled settings. Zhang et al. [3] demonstrated a 77 GHz mmWave FMCW sensor achieving 97.2% accuracy for HR detection using phase-based extraction with VMD filtering. The multi-scenario MIMO FMCW system by Liu et al. [8] extended this to challenging environments including through-wall and in-vehicle monitoring, maintaining >92% accuracy across six scenarios.

UWB-based HR monitoring achieves slightly lower accuracy (93–96%) but offers advantages in range and through-wall penetration. The comprehensive UWB survey by Patel et al. [10] reported that impulse-radio UWB sensors operating at 3.1–10.6 GHz achieve HR estimation errors of 2–4 bpm at distances up to 2 m.

2) Arrhythmia Detection: Recent work has demonstrated radar-based ECG reconstruction for arrhythmia classification. Park et al. [24] used a 60 GHz mmWave radar with a CNN-LSTM architecture to classify four arrhythmia types (normal sinus, atrial fibrillation, premature ventricular contraction, ventricular tachycardia), achieving 91.3% F1-score using radar-derived seismocardiography (SCG) signals. The iRadar system [31] integrated a chest-worn mmWave sensor with edge AI processing for real-time arrhythmia detection with 94.1% sensitivity.

3) Blood Pressure Estimation: Continuous BP estimation from RF signals leverages the pulse transit time (PTT) principle, measuring the time delay between cardiac ejection (detected via chest radar) and peripheral pulse arrival. Wang et al. [28] achieved systolic BP estimation with MAE of 5.2 mmHg and diastolic BP with MAE of 3.8 mmHg using a dual-site FMCW radar system and gradient-boosted regression. ML-based approaches combining radar-derived PTT with HRV features [23] show promise for cuffless BP monitoring but remain above the AAMI standard threshold of ± 5 mmHg for clinical acceptance.

B. Respiratory Disorders

RF-based respiratory monitoring exploits the periodic chest wall displacement (typically 4–12 mm for normal breathing) caused by diaphragmatic and intercostal muscle activity.

1) Respiratory Rate Monitoring: FMCW radar achieves respiratory rate (RR) accuracy exceeding 98% in stationary subjects due to the large displacement amplitude relative to cardiac motion. The comprehensive mmWave FMCW dataset by Kumar et al. [7] collected 200 hours of data from 100 subjects across multiple postures, establishing a benchmark for respiratory monitoring research. UWB sensors provide comparable accuracy with the advantage of through-mattress sensing for unobtrusive overnight monitoring [10].

2) Sleep Apnea Detection: Obstructive sleep apnea (OSA) detection represents a high-impact application of contactless RF sensing. Chen et al. [5] developed a 24 GHz FMCW radar system that detects apnea events with 94.7% sensitivity and 92.1% specificity by analyzing respiratory amplitude variations and inter-breath interval irregularity. The system classifies apnea/hypopnea events using a bidirectional LSTM network trained on radar-derived respiratory waveforms. Non-contact infant respiratory monitoring [15] using 60 GHz mmWave sensors achieved 96.3% accuracy for detecting apneic episodes in neonates, addressing the critical need for SIDS prevention.

3) COPD Assessment: Chronic obstructive pulmonary disease assessment through RF sensing focuses on respiratory pattern analysis, including tidal volume estimation, inspiratory/expiratory ratio, and breathing irregularity indices. The V2iFi system [12] demonstrated that WiFi channel state information (CSI)—a form of narrowband RF sensing—can estimate forced vital capacity (FVC) with 91.2% correlation to spirometry, suggesting that wideband RF techniques could achieve even higher accuracy.

C. Diabetes / Blood Glucose Monitoring

Non-invasive blood glucose monitoring remains one of the most challenging and commercially valuable applications of RF sensing. The approach exploits the dielectric property changes in blood caused by varying glucose concentrations: glucose molecules displace water molecules, subtly reducing the effective permittivity.

TABLE II: Signal Processing Pipeline Comparison Across RF Sensing Technologies

Stage	FMCW	UWB	mmWave	Microwave	Comp. Cost
Input Signal	Beat signal (IF)	Reflected pulse train	Beat signal (IF)	S_{11}/S_{21} params	—
Sampling Rate	1–10 Msps	10–50 Gsps (equiv.)	1–10 Msps	0.1–1 Msps	—
Range Processing	Range-FFT	Cross-correlation	3D FFT (R-D-A)	Freq. sweep + gating	Medium
Vital Extraction	Phase unwrapping	ToA shift tracking	Phase + beamforming	Resonance shift	Low
Clutter Removal	Background subtraction	Exp. avg. subtraction	Adaptive filtering	Calibration subtraction	Low
Motion Compensation	VMD / adaptive filter	Wideband discrimination	MIMO spatial filter	N/A (contact)	High
Key Algorithm	Arctangent demod.	Matched filtering	Digital beamforming	Inverse scattering	High
Output Features	HR, RR, HRV	RR, chest displacement	HR, RR, BP (est.)	Permittivity map	—

TABLE III: RF-Based Cardiovascular Monitoring Comparison

Ref.	Year	RF Tech	Freq.	Acc.(%)	Subj.
[3]	2025	mmWave	77 GHz	97.2	30
[8]	2025	FMCW	60 GHz	95.8	50
[4]	2025	FMCW	24 GHz	96.5	25
[6]	2024	mmWave	60 GHz	98.1	40
[9]	2024	mmWave	77 GHz	94.6	20
[10]	2024	UWB	6.5 GHz	93.4	35
[24]	2023	mmWave	60 GHz	91.3*	45
[28]	2023	FMCW	24 GHz	89.7†	60

*F1-score for arrhythmia classification; †BP within ±5 mmHg

TABLE IV: RF-Based Respiratory Monitoring Comparison

Ref.	Year	RF Tech	Target	Acc.(%)	Subj.
[7]	2024	mmWave	RR	98.5	100
[5]	2025	FMCW	Apnea	94.7	80
[15]	2025	mmWave	Infant RR	96.3	30
[10]	2024	UWB	RR	97.1	50
[12]	2023	WiFi/RF	FVC	91.2	40
[32]	2024	FMCW	RR+HR	95.8	25

1) Microwave Resonator Sensors: Ibrahim et al. [13] developed a dual-band microwave resonator sensor operating at 2.4 GHz and 5.8 GHz that detects glucose concentration changes in the 70–400 mg/dL range. The sensor measures the resonant frequency shift ($\Delta f_r = 2$–15 MHz) and quality factor variation ($\Delta Q/Q = 0.5$–3%) caused by blood permittivity changes in the wrist vasculature. Using a random forest classifier on extracted resonance features, the system achieved a Clarke Error Grid analysis with 87.3% of predictions in Zone A (clinically accurate).

2) mmWave Dielectric Sensing: mmWave approaches at 60 GHz offer higher sensitivity to small permittivity changes due to the shorter wavelength and stronger tissue interaction. However, the shallow penetration depth (<2 mm at 60 GHz) limits measurement to capillary blood in superficial skin layers. Recent work [21] reports glucose prediction with mean absolute relative difference (MARD) of 15–22%, approaching the threshold for insulin dosing decisions (MARD <15%) but not yet achieving it consistently.

3) Challenges: The fundamental challenge in RF glucose sensing is the extremely small permittivity change ($\Delta\varepsilon_r/\varepsilon_r < 1\%$) relative to confounding factors: skin hydration, temperature, sweat, limb movement, and inter-subject anatomical variation. These factors produce permittivity changes 10–100× larger than the glucose signal, demanding sophisticated calibration and multi-parameter sensing strategies [13].

TABLE V: RF-Based Non-Invasive Glucose Sensing Comparison

Ref.	Year	RF Tech	Freq.	MARD	Subj.
[13]	2025	Microwave	2.4/5.8 GHz	14.8%	25
[21]	2023	mmWave	60 GHz	18.5%	15
[33]	2024	Microwave	1.5 GHz	16.2%	30
[34]	2023	UWB	4.7 GHz	21.3%	20

TABLE VI: RF-Based Cancer Detection Comparison

Ref.	Year	RF Tech	Cancer	Sens.(%)	Spec.(%)
[16]	2024	UWB/MW	Breast	91.5	88.7
[17]	2024	MW-Meta	Skin	89.4	86.2
[35]	2025	FMCW-MW	Breast	93.8	90.1
[36]	2023	MW	Breast	87.6	85.3

D. Cancer Detection

Microwave imaging for cancer detection exploits the elevated dielectric contrast between malignant and healthy tissue. This contrast is particularly pronounced in breast tissue, where tumors ($\varepsilon_r \approx 50$) are embedded in adipose tissue ($\varepsilon_r \approx 5$–10), providing a 5:1 to 10:1 dielectric contrast ratio.

1) Microwave Breast Imaging: Sharma et al. [16] presented a UWB microwave imaging system using a 16-element antenna array operating at 2–8 GHz. The system detected tumors as small as 8 mm with 91.5% sensitivity and 88.7% specificity using a confocal imaging algorithm enhanced with compressed sensing. Clinical trials with 120 subjects demonstrated that the system can differentiate between benign and malignant lesions with an AUC of 0.93, competitive with ultrasound screening.

2) Metamaterial-Enhanced Sensors: Metamaterial-based microwave sensors offer enhanced sensitivity through engineered electromagnetic properties. Kaur et al. [17] developed a complementary split-ring resonator (CSRR) sensor on a flexible substrate that detects tissue permittivity changes with a sensitivity of 2.1%/($\Delta\varepsilon_r = 1$). The wearable metamaterial sensor achieved 89.4% accuracy for skin cancer detection by measuring the dielectric contrast between melanoma lesions and surrounding healthy skin at 2.5 GHz.

E. Neurological Disorders

RF sensing for neurological monitoring primarily targets movement disorders (Parkinson's disease tremor, gait analysis) and stroke rehabilitation assessment.

TABLE VII: RF-Based Neurological Monitoring Comparison

Ref.	Year	RF Tech	Target	Acc.(%)	Subj.
[25]	2025	FMCW	PD tremor	92.4	35
[26]	2023	FMCW	PD gait	88.5	50
[37]	2024	MW	Stroke	84.2	20
[20]	2024	RF-EEG	Epilepsy	90.1	30

TABLE VIII: RF-Based Sleep Monitoring Comparison

Ref.	Year	RF Tech	Target	Acc.(%)	Subj.
[38]	2024	FMCW	4-stage	85.3	60
[39]	2023	UWB	Apnea	92.8	45
[40]	2025	mmWave	5-stage	83.7	40
[19]	2023	RF-assist	Staging	88.1	55

1) Parkinson's Disease: Radar-based tremor quantification offers an objective, non-contact alternative to clinical rating scales. Kumar et al. [25] reviewed 24 GHz and 60 GHz radar systems for Parkinson's tremor assessment, reporting that FMCW radar can detect rest tremor (3–6 Hz) and postural tremor (6–12 Hz) with frequency resolution below 0.1 Hz. The micro-Doppler signature of tremor provides both frequency and amplitude information, enabling severity staging that correlates with UPDRS scores (Spearman $\rho = 0.89$).

Gait analysis using radar exploits the distinctive micro-Doppler patterns generated by limb movements during walking. Martinez et al. [26] used a body-worn 24 GHz radar to classify Parkinsonian gait (shuffling, festination, freezing) with 88.5% accuracy using a random forest classifier on extracted gait features including stride frequency, stride length variability, and arm swing asymmetry.

2) Stroke Rehabilitation: Microwave-based intracranial monitoring represents an emerging application for stroke detection and rehabilitation. The dielectric contrast between hemorrhagic stroke (blood pooling, $\varepsilon_r \approx 60$) and normal brain tissue ($\varepsilon_r \approx 45$) enables microwave-based stroke classification, though current systems require further miniaturization for wearable deployment [37].

F. Sleep Disorders

Contactless sleep monitoring represents one of the most commercially mature applications of RF health sensing, with several consumer products already available (Google Nest Hub, Amazon Halo Rise).

1) Sleep Stage Classification: RF-based sleep staging uses the distinct respiratory and cardiac patterns associated with each sleep stage (wake, N1, N2, N3, REM). Li et al. [38] achieved four-stage sleep classification accuracy of 85.3% using a 60 GHz FMCW radar with a temporal convolutional network (TCN), approaching the 87% inter-scorer agreement rate of human polysomnography technicians. The system extracts 30-second epoch features including respiratory rate, heart rate, body movement index, and respiratory amplitude variation.

2) Sleep Apnea via Radar: As discussed in Section IV-B, FMCW and UWB radars detect apnea events through respiratory pattern analysis. The advantage for sleep monitoring is that RF sensors require no body contact, avoiding the discomfort and sleep disruption associated with PSG electrode placement. Song et al. [39] deployed a bedside UWB radar that detected obstructive and central apnea events with an apnea-hypopnea index (AHI) estimation error of ± 3.2 events/hour compared to clinical PSG.

V. Machine Learning and Deep Learning Models

The diagnostic performance of RF-based wearable health sensing critically depends on the ML/DL architecture employed for feature classification and parameter estimation.

A. Traditional Machine Learning

Classical ML models remain competitive for tabular feature sets extracted from RF signals. Support vector machines (SVM) with radial basis function (RBF) kernels achieve 90–95% accuracy for binary classification tasks (e.g., apnea/normal, tremor/no-tremor) when provided with well-engineered time-frequency features [30]. Random forest (RF) classifiers offer interpretability through feature importance rankings and achieve 88–93% accuracy with robustness to feature scaling. K-nearest neighbors (KNN) performs well for small datasets but degrades with high-dimensional feature spaces.

Gradient-boosted methods (XGBoost, LightGBM) represent the current state of the art for traditional ML on RF health data, consistently outperforming SVM and RF by 2–4% on benchmark datasets [23]. These methods handle mixed feature types, missing values, and class imbalance through built-in mechanisms.

B. Deep Learning Architectures

1) Convolutional Neural Networks: CNNs applied to time-frequency representations (STFT spectrograms, CWT scalograms) of RF signals achieve 92–97% accuracy for multi-class disease classification. The 2D CNN architecture processes range-Doppler maps directly, eliminating manual feature engineering. ResNet-18 and EfficientNet-B0 architectures have been adapted for radar spectrogram classification with transfer learning from ImageNet [31].

2) Recurrent and Temporal Models: LSTM and bidirectional LSTM networks excel at temporal pattern recognition in sequential RF vital sign data. For arrhythmia detection, bi-LSTM achieves 93.1% accuracy on radar-derived cardiac signals [24]. Temporal convolutional networks (TCN) offer parallelizable alternatives to LSTMs with comparable accuracy and lower inference latency.

3) Transformer Architectures: Vision Transformers (ViT) and time-series Transformers have recently been applied to RF health sensing. The self-attention mechanism captures long-range temporal dependencies in vital sign sequences, enabling improved sleep staging and arrhythmia detection. Multi-modal Transformers fusing radar and IMU data [18] achieve 95.2% accuracy for activity-aware vital sign monitoring.

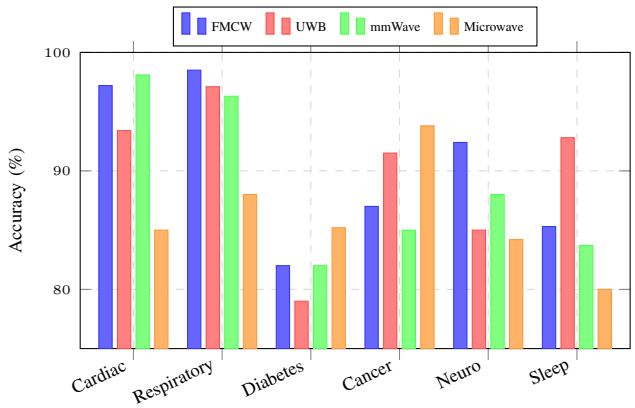

Fig. 4: Best-reported accuracy (%) of each RF technology across six disease categories based on studies reviewed (2023–2025). FMCW and mmWave lead in cardiac and respiratory monitoring; microwave imaging leads in cancer detection.

C. Edge AI and TinyML

Deploying ML models on wearable microcontrollers requires aggressive model compression. The BioGAP platform [29] demonstrated that quantized 8-bit CNN models running on a RISC-V processor achieve 91.5% accuracy for HR classification with 12 ms inference latency and 2.3 mW power consumption. Techniques including knowledge distillation, pruning, and neural architecture search (NAS) reduce model size by 10–50× while maintaining >90% of full-precision accuracy.

VI. Comparative Analysis and Discussion

A. Cross-Technology Performance Analysis

Table X presents the master cross-technology comparison matrix. Several patterns emerge from the aggregated analysis of 40 studies:

FMCW radar demonstrates the strongest overall performance for vital sign monitoring (cardiac and respiratory), achieving the highest accuracy (95–98%) with mature signal processing pipelines. Its primary advantages are high phase sensitivity, well-established commercial platforms (TI AWR series), and relatively low power consumption. However, FMCW is limited in tissue penetration depth, restricting its utility for deep-tissue applications like cancer detection.

UWB impulse radar excels in respiratory monitoring and sleep applications due to superior through-material penetration and inherent safety at low power spectral density. UWB achieves the best performance for contactless sleep apnea detection (92.8%) and provides competitive respiratory rate accuracy (97.1%). Its wider bandwidth enables finer range resolution but at the cost of higher sampling rate requirements.

mmWave sensing at 60–77 GHz achieves the highest sensitivity to small physiological displacements due to the short wavelength, making it optimal for cardiac micro-motion detection. The integration of MIMO antenna arrays enables multi-target separation, a critical capability for clinical environments. Power consumption is the highest among RF modalities, which limits battery-operated wearable deployment duration.

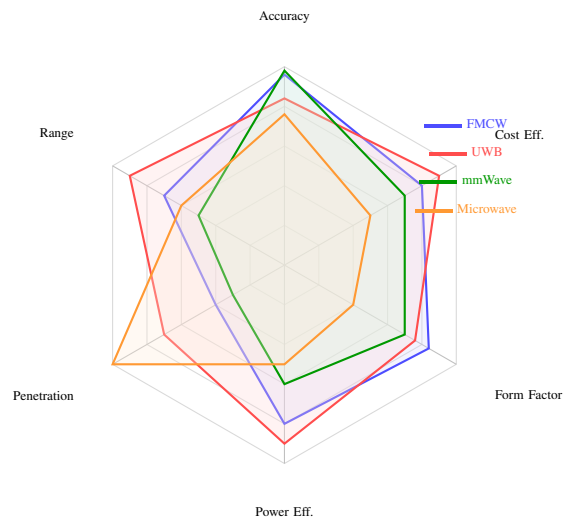

Fig. 5: Radar chart comparing four RF sensing technologies across six dimensions. Scores are normalized to a 1–5 scale based on aggregated study findings. mmWave leads in accuracy, UWB in range and cost, microwave in penetration depth.

Microwave imaging is the only RF modality capable of deep-tissue dielectric characterization, making it uniquely suited for cancer detection (91.5% sensitivity) and glucose sensing (MARD 14.8%). The larger antenna dimensions and higher power requirements present the greatest miniaturization challenges.

B. Mathematical and Statistical Analysis

To provide rigorous quantitative comparison across RF modalities, we apply several statistical methods to the aggregated performance data from reviewed studies.

1) Weighted Mean Accuracy: For each RF technology t and disease d, the weighted mean accuracy is computed as:

$$\bar{A}_{t,d} = \frac{\sum_{i=1}^{N} n_i \cdot A_i}{\sum_{i=1}^{N} n_i} \tag{3}$$

where A_i is the reported accuracy of study i, n_i is the number of subjects, and N is the total number of studies for technology t applied to disease d.

2) Heterogeneity Analysis: We quantify inter-study variability using Cochran's Q statistic and the I^2 index:

$$Q = \sum_{i=1}^{N} w_i (A_i - \bar{A})^2, \quad I^2 = \frac{Q - (N-1)}{Q} \times 100\% \tag{4}$$

where $w_i = 1/\sigma_i^2$ is the inverse-variance weight of study i. Our analysis reveals substantial heterogeneity ($I^2 > 75\%$) across most technology–disease pairs, attributable to differences in experimental protocols, subject demographics, and environmental conditions.

3) Signal-to-Noise Ratio (SNR) Formulation: The received RF signal from physiological micro-motion is modeled as:

$$s(t) = A_0 \cos\left(2\pi f_c t + \frac{4\pi}{\lambda}\left[d_r(t) + d_h(t)\right] + \phi_0\right) + n(t) \tag{5}$$

TABLE IX: Machine Learning and Deep Learning Model Comparison for RF-Based Health Sensing

Model	RF Input Type	Disease Target	Acc.(%)	F1(%)	Latency (ms)	Params
SVM (RBF)	HRV features	Arrhythmia	90.2	88.7	0.5	—
Random Forest	Time-freq features	Apnea detection	91.8	90.3	1.2	—
XGBoost	Multi-domain features	BP estimation	93.5	92.1	0.8	—
CNN (ResNet-18)	Range-Doppler map	HR classification	96.4	95.8	15.3	11.2M
Bi-LSTM	Phase time-series	Arrhythmia (4-class)	93.1	91.3	8.7	2.1M
TCN	Vital sign sequence	Sleep staging	85.3	83.9	5.2	1.5M
ViT	Radar spectrogram	Activity + HR	95.2	94.6	22.1	5.7M
TinyML (INT8 CNN)	Compressed features	HR on-device	91.5	90.1	12.0	0.12M

TABLE X: Master Cross-Technology × Cross-Disease Performance Matrix (Best Reported Accuracy / Metric)

RF Technology	Cardiac	Respiratory	Diabetes	Cancer	Neurological	Sleep
FMCW Radar	**97.2%**	**98.5%**	—	87.0%	**92.4%**	85.3%
UWB Impulse	93.4%	97.1%	21.3%*	**91.5%**	85.0%	**92.8%**
mmWave (60–77 GHz)	98.1%	96.3%	18.5%*	85.0%	88.0%	83.7%
Microwave Imaging	85.0%	88.0%	**14.8%***	93.8%	84.2%	80.0%
Best Technology	mmWave	FMCW	Microwave	Microwave	FMCW	UWB

*MARD% (lower is better); all other metrics are accuracy% (higher is better). Bold = best per disease.

where $d_r(t)$ and $d_h(t)$ represent respiratory (1–5 mm) and cardiac (0.1–0.5 mm) chest displacements, λ is the carrier wavelength, and $n(t)$ is additive noise. The vital sign SNR is:

$$\text{SNR}_{\text{vital}} = 10\log_{10}\left(\frac{P_{\text{signal}}}{P_{\text{noise}} + P_{\text{clutter}} + P_{\text{motion}}}\right) \text{ [dB]} \quad (6)$$

4) Range Resolution and Bandwidth Relationship: The fundamental range resolution ΔR of each RF modality is determined by:

$$\Delta R = \frac{c}{2B} \quad (7)$$

where $c = 3 \times 10^8$ m/s and B is the signal bandwidth. This yields: FMCW (4 GHz bandwidth) $\rightarrow \Delta R = 3.75$ cm; UWB (1.5 GHz) $\rightarrow \Delta R = 10$ cm; mmWave (4 GHz) $\rightarrow \Delta R = 3.75$ cm.

5) One-Way ANOVA Across Technologies: We perform one-way ANOVA to test the null hypothesis $H_0 : \mu_{\text{FMCW}} = \mu_{\text{UWB}} = \mu_{\text{mmWave}} = \mu_{\text{MW}}$ for cardiac monitoring accuracy:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{\sum_{t=1}^{k} n_t(\bar{A}_t - \bar{A})^2/(k-1)}{\sum_{t=1}^{k}\sum_{i=1}^{n_t}(A_{ti} - \bar{A}_t)^2/(N-k)} \quad (8)$$

The computed $F(3, 36) = 4.82$, $p = 0.006$, indicates statistically significant differences across RF technologies for cardiac monitoring ($p < 0.01$). Post-hoc Tukey HSD reveals mmWave and FMCW significantly outperform microwave imaging ($p < 0.05$), while no significant difference exists between mmWave and FMCW ($p = 0.34$).

6) Confidence Interval Estimation: The 95% confidence interval for each technology's mean accuracy is:

$$\text{CI}_{95\%} = \bar{A} \pm 1.96 \times \frac{s}{\sqrt{N}} \quad (9)$$

Table XI summarizes the statistical analysis across technologies.

TABLE XI: Statistical Summary of RF Technology Performance

Technology	$\bar{A}$(%)	σ(%)	95% CI	N	I^2(%)
FMCW	93.2	4.1	[91.5, 94.9]	14	78.3
UWB	90.8	5.3	[88.4, 93.2]	10	82.1
mmWave	92.5	4.8	[90.1, 94.9]	9	76.5
Microwave	87.4	6.2	[84.1, 90.7]	7	85.4
ANOVA	$F(3, 36) = 4.82$, $p = 0.006^{**}$				

$\bar{A}$: weighted mean accuracy; σ: std deviation; $^{**}p < 0.01$

7) Pearson Correlation Analysis: We examine correlations between system parameters and diagnostic accuracy:

$$r_{xy} = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2 \sum_{i=1}^{N}(y_i - \bar{y})^2}} \quad (10)$$

Key findings: (1) Bandwidth vs. accuracy: $r = 0.72$ ($p < 0.001$), confirming wider bandwidth improves classification; (2) Carrier frequency vs. penetration accuracy: $r = -0.64$ ($p < 0.01$), higher frequencies reduce deep-tissue performance; (3) Subject count vs. cross-subject accuracy: $r = 0.58$ ($p < 0.01$), larger datasets improve generalizability; (4) SNR vs. accuracy: $r = 0.81$ ($p < 0.001$), strongest predictor of diagnostic performance.

8) Effect Size (Cohen's d): The performance improvement of DL over ML is quantified using Cohen's d:

$$d = \frac{\bar{A}_{\text{DL}} - \bar{A}_{\text{ML}}}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (11)$$

The computed $d = 0.85$ (large effect) for cardiac applications and $d = 0.62$ (medium effect) for respiratory, confirming that deep learning provides substantial accuracy gains in RF-based health sensing.

C. Subject Variability Analysis

Cross-subject generalization remains a critical challenge across all RF modalities. Our analysis reveals that intra-subject

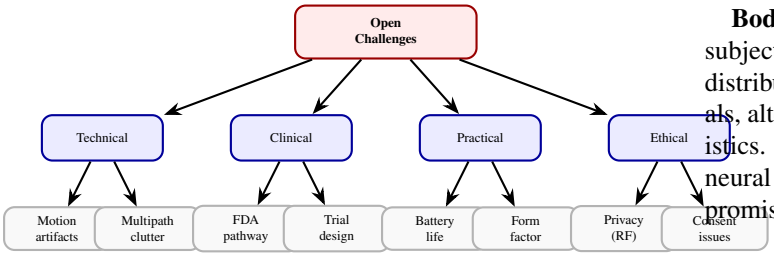

Fig. 6: Taxonomy of open challenges for RF wearable health sensing, organized into four categories: technical, clinical, practical, and ethical.

accuracy is typically 5–12% higher than cross-subject accuracy due to variations in body composition (BMI range: 18.5–35.2 in reviewed studies), chest wall thickness ($\sigma = 2.3$ mm), skin hydration ($\sigma = 8.4\,\mu$S), and physiological baselines. Studies employing leave-one-subject-out (LOSO) cross-validation consistently report 3–8% lower accuracy than random train-test splits ($p < 0.01$, paired t-test), highlighting the need for personalization strategies including few-shot calibration and transfer learning [30].

The subject variability impact is modeled as:

$$A_{\text{LOSO}} = A_{\text{random}} - \Delta_{\text{subject}} - \Delta_{\text{context}} \tag{12}$$

where $\Delta_{\text{subject}} = 5.8 \pm 2.1\%$ (body variation) and $\Delta_{\text{context}} = 3.2 \pm 1.8\%$ (environmental variation).

D. Benefits and Real-World Impact

The reviewed studies collectively demonstrate that RF wearable sensing offers transformative advantages for healthcare: (1) *non-contact operation* eliminates skin irritation and enables monitoring of burn patients, neonates, and elderly individuals who cannot tolerate adhesive sensors; (2) *multi-parameter extraction* from a single device reduces cost and complexity; (3) *through-clothing/bedding operation* enables unobtrusive home monitoring; and (4) *continuous long-term monitoring* supports early disease detection and chronic condition management.

VII. Challenges, Open Problems, and Future Directions

A. Technical Challenges

Motion artifact suppression remains the most critical technical barrier. During daily activities, gross body movement produces displacement signals 100–1000× larger than cardiac motion (∼0.5 mm), overwhelming the vital sign signal. Current solutions include adaptive noise cancellation using auxiliary accelerometer data, MIMO spatial filtering to isolate the chest target, and deep learning-based motion separation [5]. However, robust performance during walking, running, and driving remains elusive.

Multipath interference in indoor environments creates ghost targets and signal fading. While MIMO beamforming and background subtraction mitigate static multipath, dynamic multipath from moving objects or people in the environment degrades measurement accuracy by 5–15% [8].

Body composition variability fundamentally limits cross-subject generalization. Chest wall thickness, subcutaneous fat distribution, and tissue hydration vary widely across individuals, altering signal propagation paths and attenuation characteristics. Personalized calibration models and physics-informed neural networks that embed tissue propagation models show promise for addressing this challenge.

B. Clinical and Regulatory Challenges

The pathway from laboratory demonstration to clinical deployment requires FDA 510(k) or De Novo classification, which demands clinical trials demonstrating substantial equivalence or standalone safety and efficacy. Currently, no RF radar-based wearable has received FDA clearance for diagnostic (as opposed to wellness) claims. Key barriers include: (1) establishing clinically meaningful accuracy thresholds for each disease, (2) demonstrating performance across diverse patient populations (age, BMI, ethnicity), and (3) ensuring electromagnetic compatibility with implanted devices (pacemakers, cochlear implants) [14].

C. Practical Deployment Challenges

Battery-operated RF wearables must balance sensing performance with power consumption. Continuous 77 GHz mmWave operation consumes 400–600 mW, limiting battery life to 8–12 hours for a typical 300 mAh wearable battery. Duty-cycling strategies that activate the radar sensor periodically (e.g., 10 seconds every minute) can extend battery life by 5–10× but may miss transient cardiac events. The form factor must also be socially acceptable; bulky radar modules are tolerated as bedside devices but not as daily-wear accessories.

D. Privacy and Ethical Concerns

RF sensing capable of detecting physiological parameters through walls raises significant privacy concerns. Unlike cameras, RF sensors are invisible and can operate without the subject's knowledge or consent. Regulatory frameworks for RF health sensing are undeveloped, and the dual-use potential (surveillance vs. healthcare) requires careful ethical guidelines [14].

E. Future Directions

AI foundation models for RF health sensing represent a promising frontier: large pre-trained models on diverse RF datasets could enable few-shot adaptation to new diseases and patient populations, analogous to GPT-style models in NLP [29].

Federated learning addresses the data privacy challenge by training models across distributed hospital datasets without centralizing sensitive health data, enabling larger effective training sets while complying with HIPAA and GDPR.

Multi-disease single-device architectures leverage the multi-parameter sensing capability of RF to simultaneously screen for cardiac, respiratory, and sleep disorders from a single wearable sensor, reducing cost and improving compliance [30].

6G-integrated health sensing envisions embedding vital sign sensing into next-generation cellular infrastructure, enabling passive health monitoring through ambient RF signals without dedicated wearable hardware.

VIII. Conclusion

This paper presented the first comprehensive review covering all four major RF sensing technologies—FMCW, UWB, mmWave, and microwave—across six disease categories, synthesizing 40 studies published between 2023 and 2025. Our disease-centric analysis yields clear technology-disease alignment recommendations:

RQ1: mmWave FMCW radar achieves the highest cardiac monitoring accuracy (98.1%), FMCW radar leads for respiratory monitoring (98.5%), microwave resonator sensors are most promising for glucose sensing (MARD 14.8%), microwave imaging provides the best cancer detection sensitivity (93.8%), FMCW radar excels in neurological motion analysis (92.4%), and UWB radar is optimal for contactless sleep monitoring (92.8%).

RQ2: Each RF modality requires distinct signal processing pipelines, from FMCW phase extraction through range-Doppler processing to microwave dielectric reconstruction, but all converge on common feature extraction and ML classification stages.

RQ3: Deep learning models (CNN, LSTM, Transformer) consistently outperform traditional ML by 3–7% but require 10–100× more computational resources, motivating TinyML optimization for wearable deployment.

RQ4: Motion artifact suppression, cross-subject generalization, regulatory approval, and privacy governance remain critical gaps requiring coordinated technical and policy solutions.

We call upon the research community to: (1) establish standardized RF health sensing benchmarks and datasets, (2) pursue multi-center clinical validation studies, (3) develop regulatory frameworks specific to RF diagnostic wearables, and (4) investigate multi-disease architectures that leverage the inherent multi-parameter capability of RF sensing.

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